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BigMart Sales Analysis

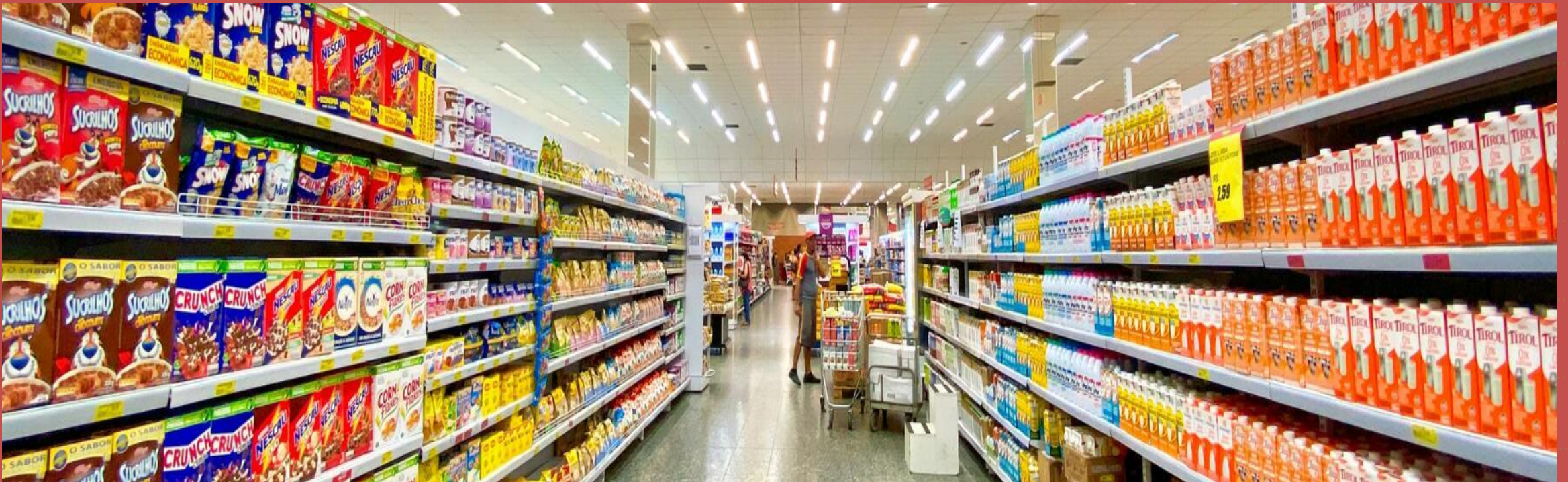


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Business Overview

This project analyzes **BigMart** sales data to uncover actionable insights into product performance, store efficiency, and category sales trends across different regions, enabling data-driven decisions to improve overall revenue and inventory management.



Data Sources & Description

- The dataset, sourced from **Kaggle's Big Mart Sales dataset**, contains **8,523 records** and **12 variables** describing **1,559 products** sold across **10 stores** located in different cities.
- It includes detailed attributes of both products and stores, such as **item weight, visibility, category, outlet type, and establishment year**, providing a comprehensive foundation for sales performance analysis.

Big Mart Sales

42

Code

Download

Data Card

Code (31)

Discussion (1)

Suggestions (0)

train_v9rqX0R.csv (869.54 kB)

Detail

Compact

Column

10 of 12 columns

Item_Identifier Unique product ID	Item_Weight Weight of product	Item_Fat_Content Whether the product is low fat or not	Item_Visibility The % of total display area of all products in a store allocated to the particular product	Item_Type The category to which the product belongs	Item_Outlet_Identifier The outlet where the product is sold
1559 unique values	 4.55 21.4	Low Fat 60% Regular 34% Other (545) 6%	 0 0.33	Fruits and Vegetables 14% Snack Foods 14% Other (6091) 71%	Maxi Center 31.3% Mini Supermarket 14.1% Supermarket 18.2% Household 53.8%
FDN15	17.5	Low Fat	0.016768875	Meat	141.1
FDX07	19.2	Regular	0	Fruits and Vegetables	182.1
NCD19	8.93	Low Fat	0	Household	53.8

Data Explorer

Version 1 (869.54 kB)

train_v9rqX0R.csv

Summary

1 file

12 columns

Data Sources & Description

Attribute	Description	Type / Purpose
Item_Identifier	Unique product code identifying each item	ID / Categorical
Item_Weight	Weight of the product (in kilograms)	Numerical
Item_Fat_Content	Indicates whether the product is low-fat or regular	Categorical
Item_Visibility	% of total shelf display area allocated to the product	Numerical
Item_Type	Product category or type	Categorical
Item_MRP	Maximum Retail Price (list price)	Numerical
Outlet_Identifier	Unique store ID	ID / Categorical
Outlet_Establishment_Year	Year the outlet was established	Numerical / Time
Outlet_Size	Physical size of the store (e.g., Small, Medium, Large)	Categorical
Outlet_Location_Type	City type where the store is located (Urban, Tier 2, etc.)	Categorical
Outlet_Type	Store format (Grocery Store, Supermarket Type 1/2/3)	Categorical
Item_Outlet_Sales	Total sales of the item in the specific outlet	Target / Numerical

Data Wrangling & Preparation

Step	Action Performed	Purpose / Outcome
1	Used COUNTBLANK () to identify missing values – 1,463 in <i>Item_Weight</i> and 2,410 in <i>Outlet_Size</i> .	Detect missing data.
2	Used COUNTIF () to count zero values in <i>Item_Visibility</i> .	Identify invalid entries.
3	Filled missing <i>Item_Weight</i> values with the average weight for each <i>Item_Identifier</i> .	Maintain consistency by imputing missing values logically.
4	Replaced zero values in <i>Item_Visibility</i> with the mean visibility.	Correct unrealistic zero entries.
5	Removed <i>Outlet_Size</i> column due to many nulls and low analytical importance.	Simplify dataset and improve model quality.
6	Replaced "lf" with "Low Fat" in <i>Item_Fat_Content</i> .	Standardize categorical values.
7	Replaced "reg" with "Regular" in <i>Item_Fat_Content</i> .	Ensure consistent data formatting.
8	Converted <i>Item_MRP</i> and <i>Item_Outlet_Sales</i> columns to currency format.	Enhance readability and reporting clarity.

Business Challenges

This analysis aims to address key business questions to understand sales performance and market trends across product categories and store locations:

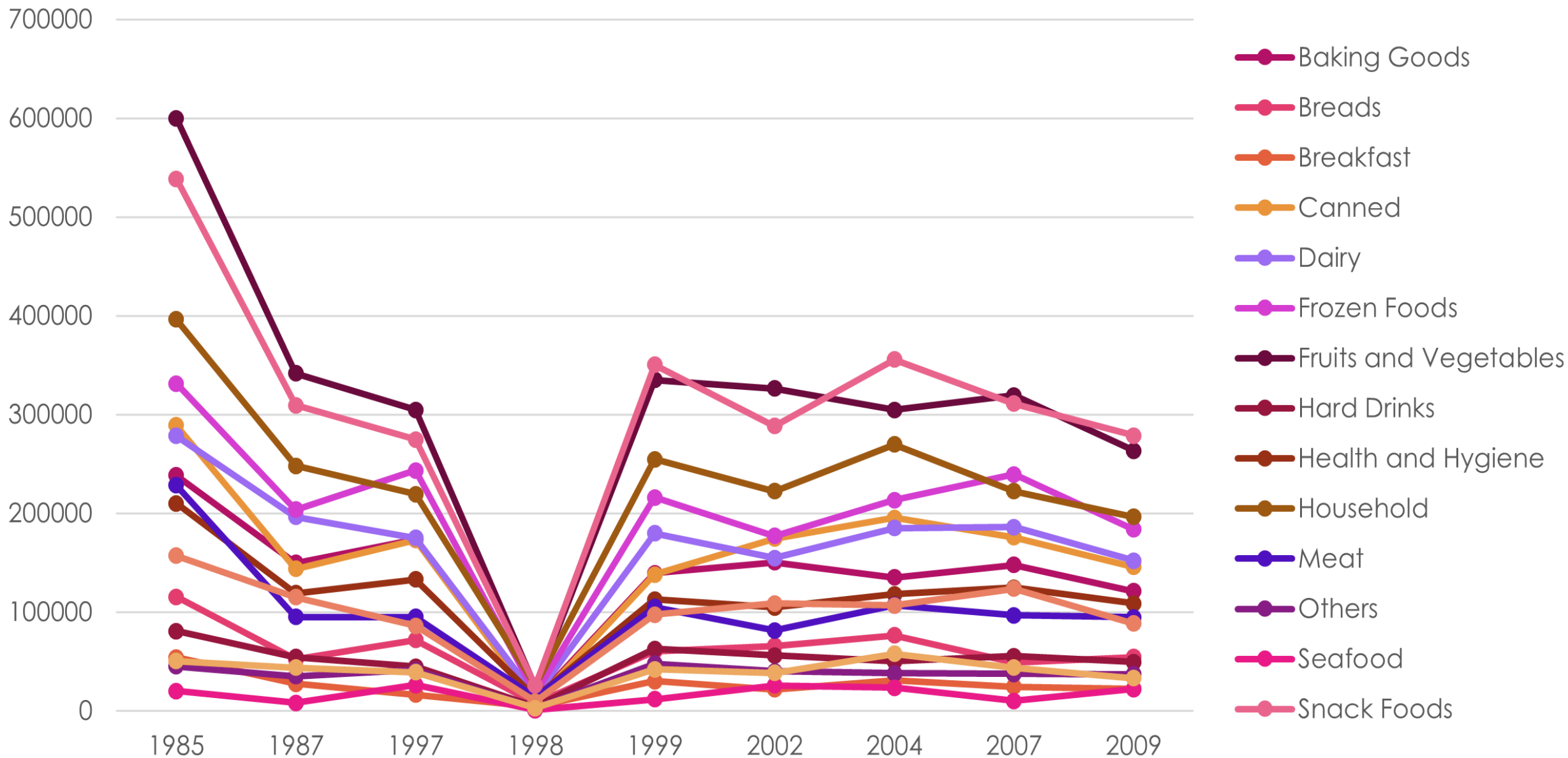
- **Annual Sales by Category** – Identify which product categories drive the highest annual sales.
- **Sales Drivers** – Determine the main factors influencing sales performance (e.g., pricing, product category, locations).
- **Price-Sales Relationship** – Measure the **correlation** between product prices and total sales.
- **Category Pricing by Location** – Calculate the **average product price** of each category across different locations.
- **Sales Distribution by Location** – Evaluate the **total category sales** across city types or regions.

Analytical Approach & Solutions

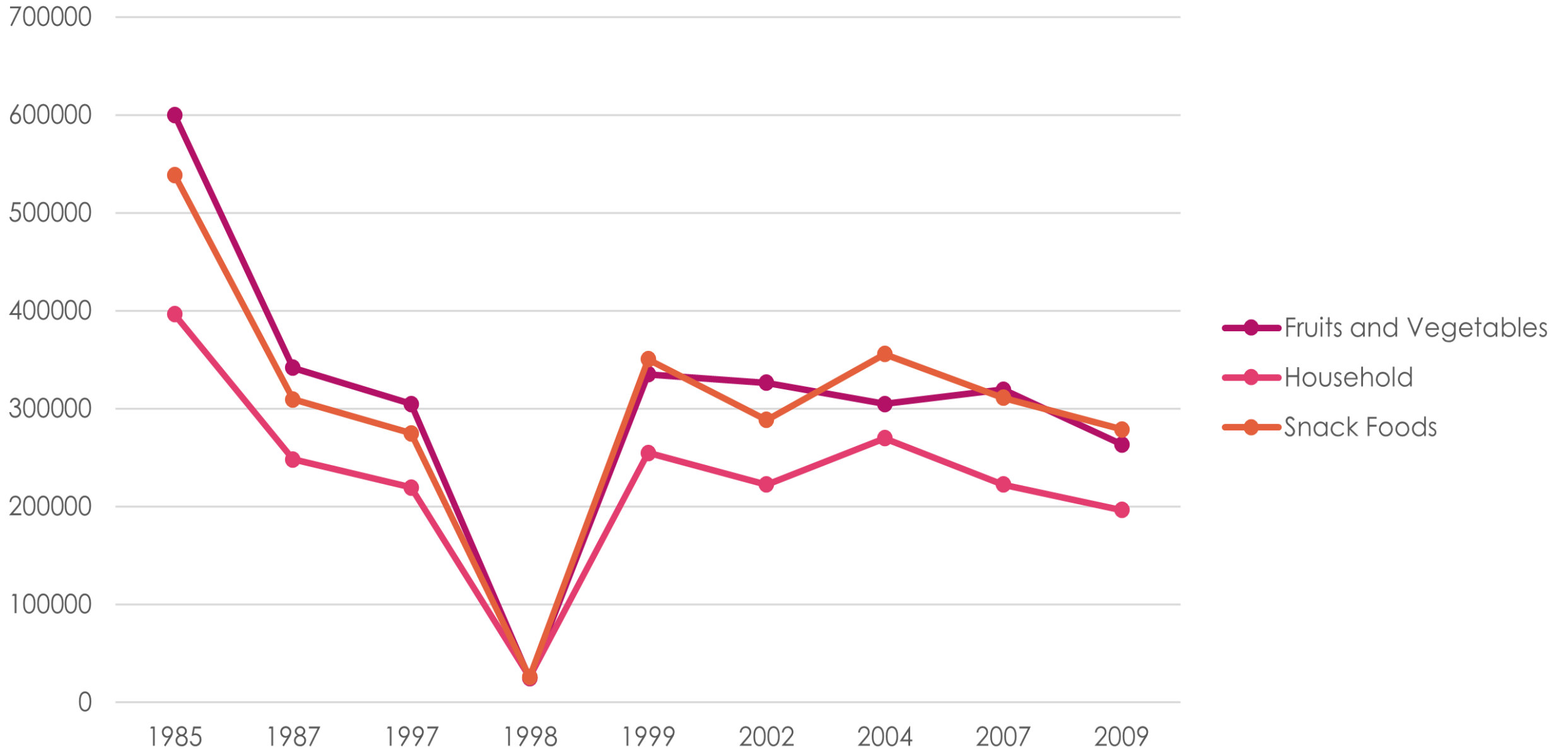
To address the business challenges, a structured analytical workflow was followed using Excel for data analysis and visualization:

- **Exploratory Data Analysis (EDA):** Used pivot tables and descriptive statistics to understand sales distribution by category, and region. Identified outliers and key trends across product types and locations.
- **Sales Trend Analysis:** Calculated annual sales per category using SUMIFS and visualized results with line charts. Analyzed year-over-year outlet growth to identify expansion patterns.
- **Correlation & Price Impact:** Used CORREL function to measure the relationship between product prices (MRP) and sales performance.
- **Regional & Category Insights:** Computed average product prices and total sales per category per location using AVERAGEIFS and SUMIFS. Visualized sales distribution using clustered bars to compare performance across cities.
- **Business Solutions Proposed:** Recommended focusing on high-performing categories and optimizing pricing for regions with low sales correlation. Suggested improving visibility for underperforming items to enhance sales consistency.

Sum of item outlet sales per year



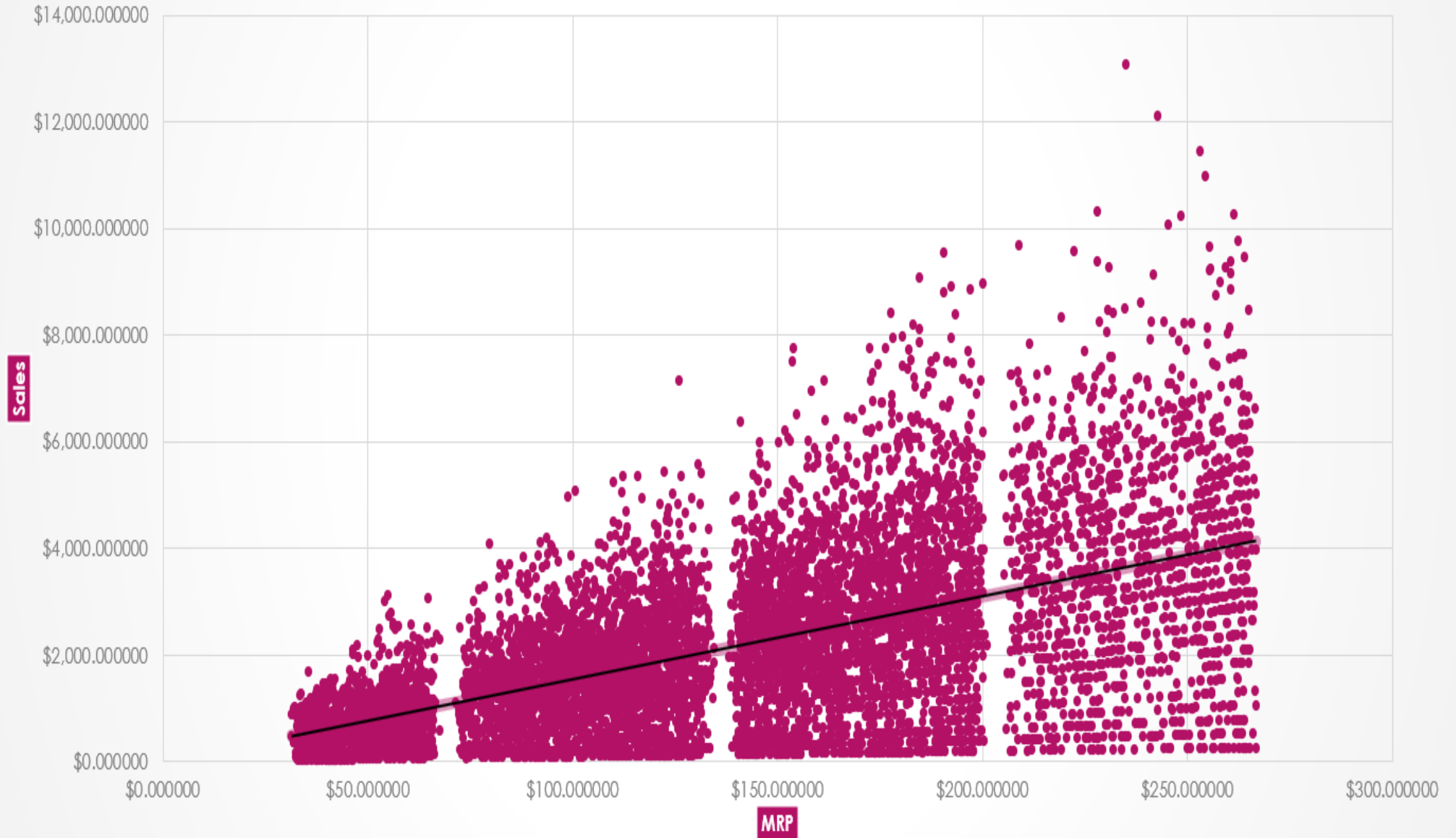
Sum of item outlet sales per year



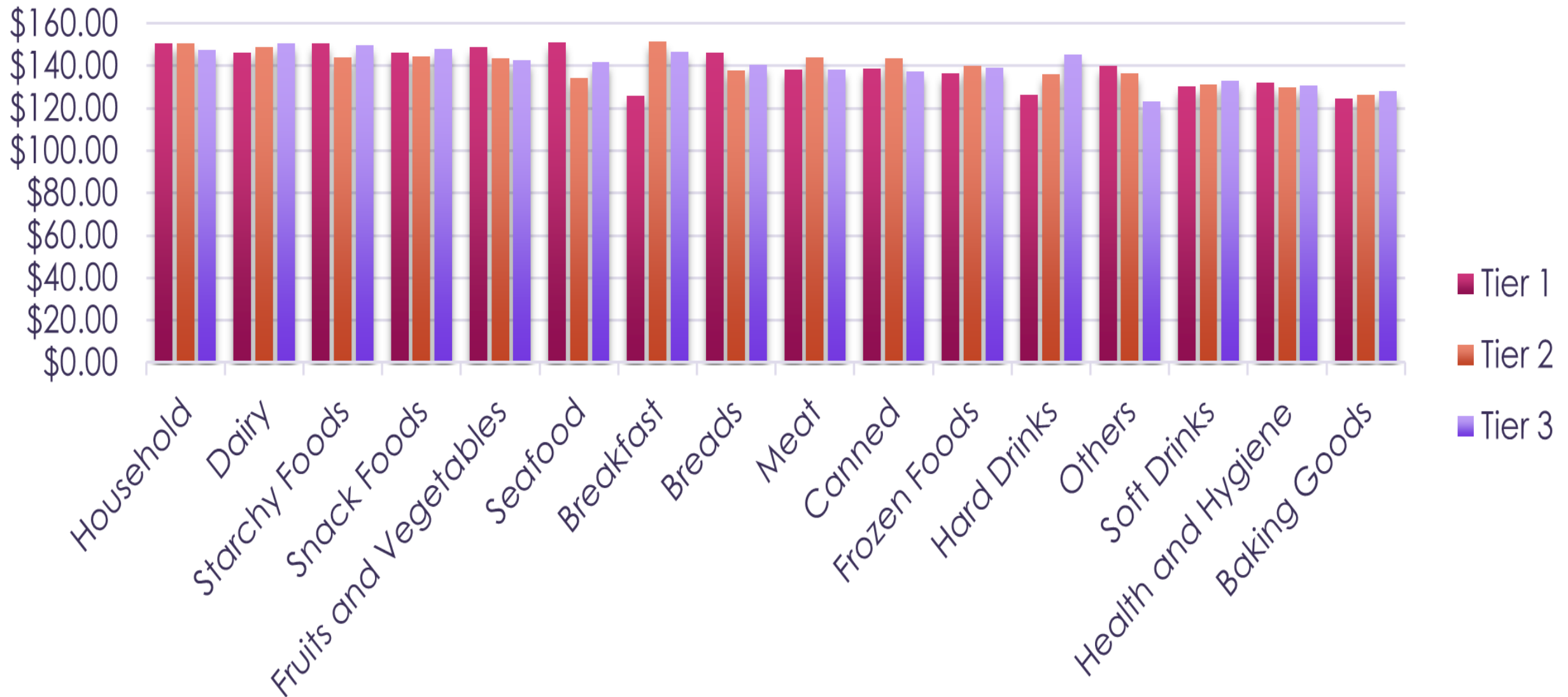
Correlation

0.567574447

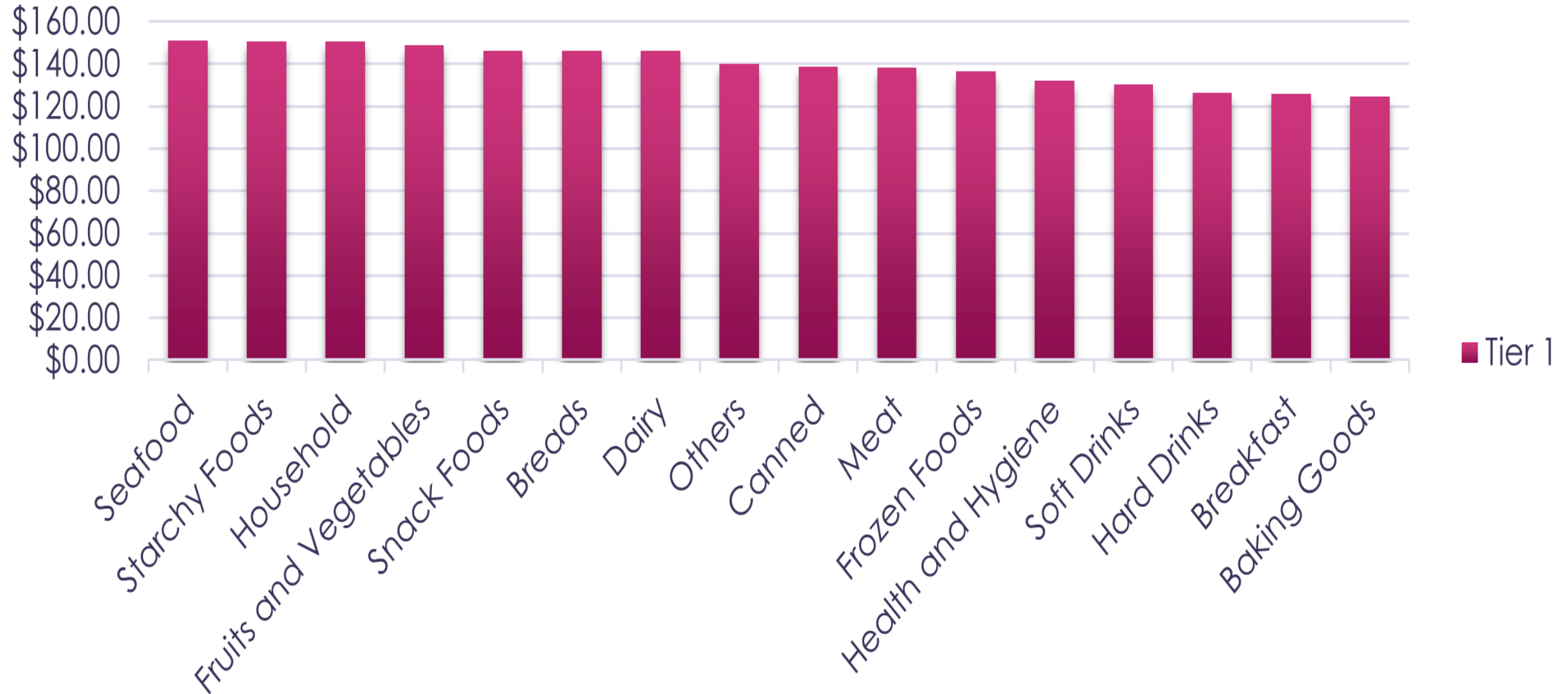
Sales and MRP relationship



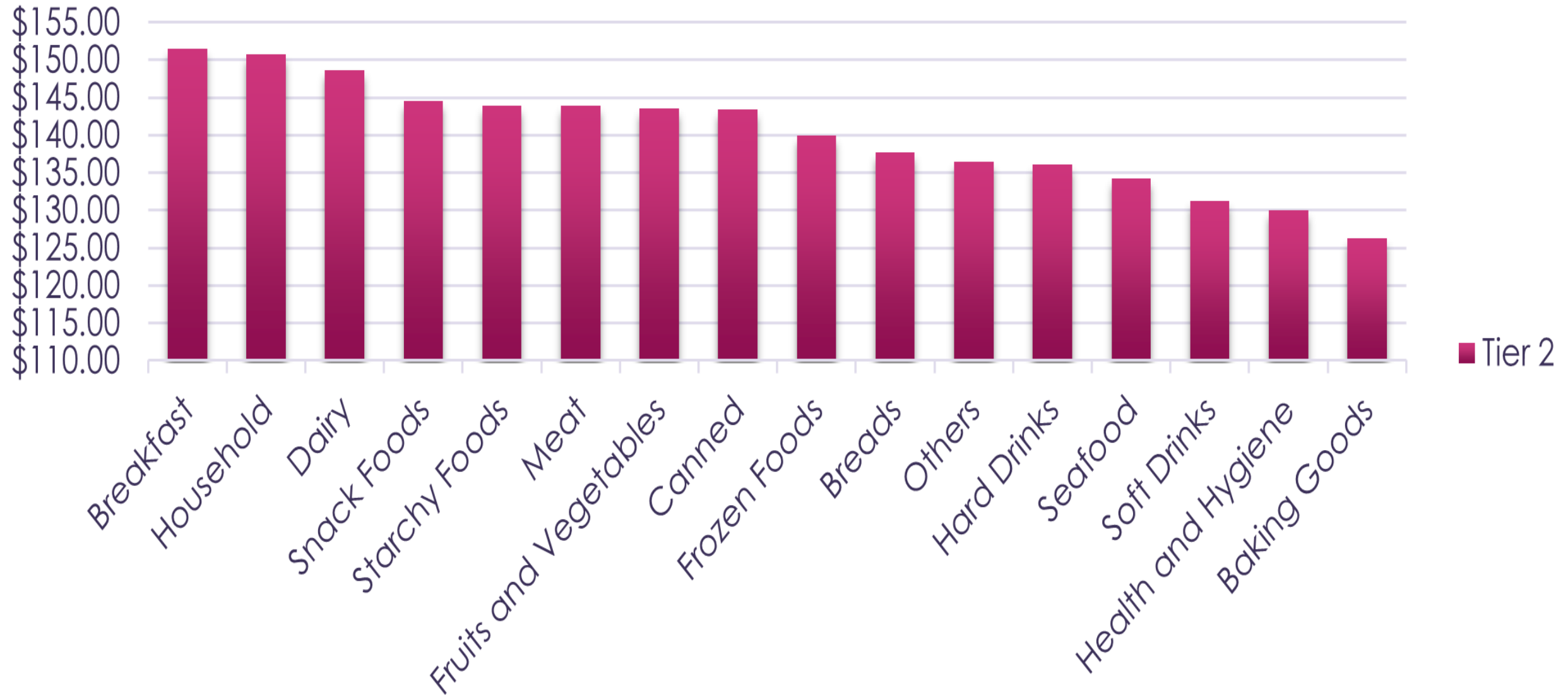
Average items MRP per category in each tier



Average items MRP per category in each tier

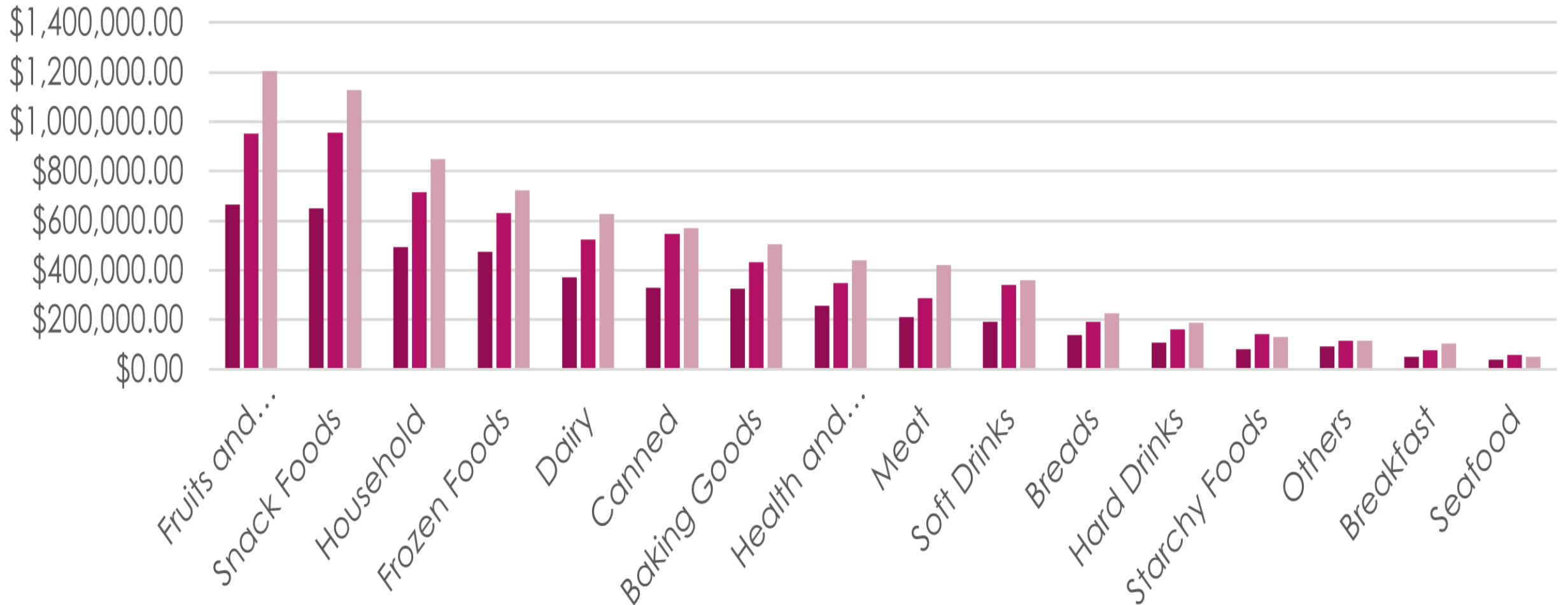


Average items MRP per category in each tier



Total sales for each category in each tier

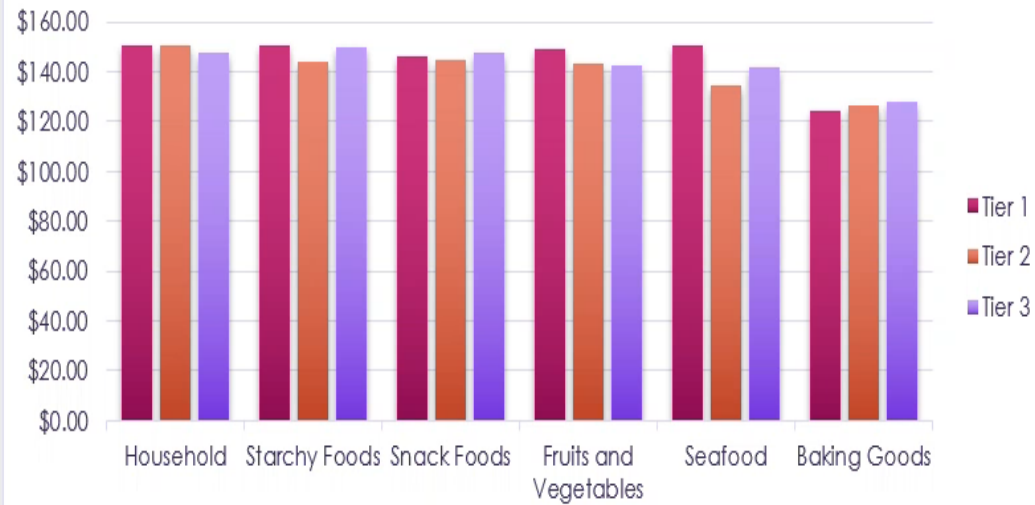
Tier 1 Tier 2 Tier 3



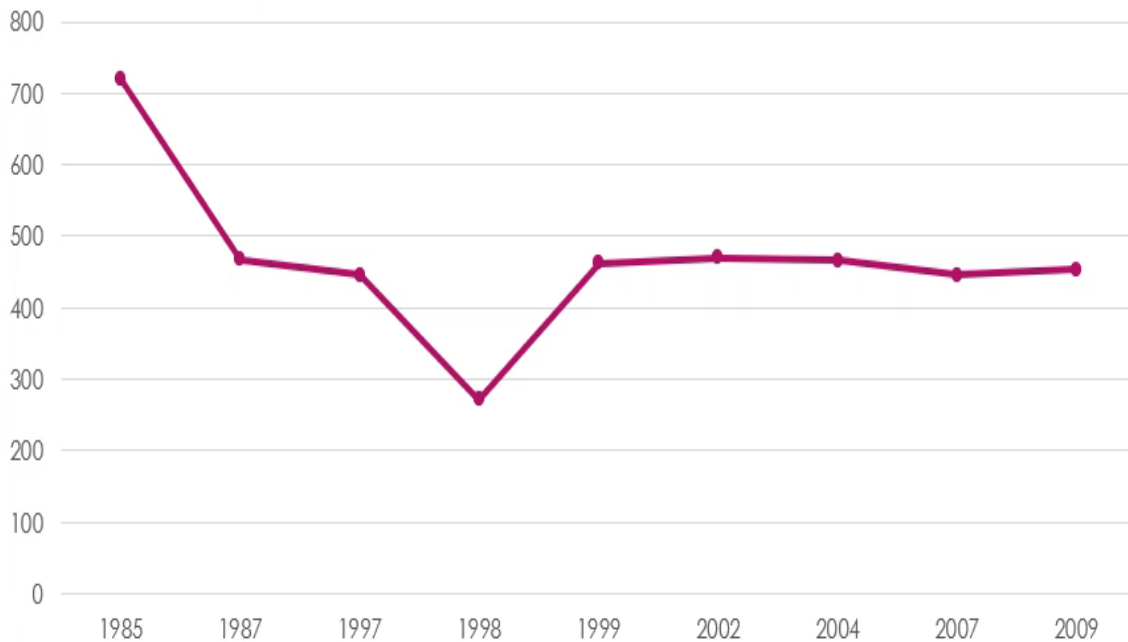
Total sales for each category in each tier



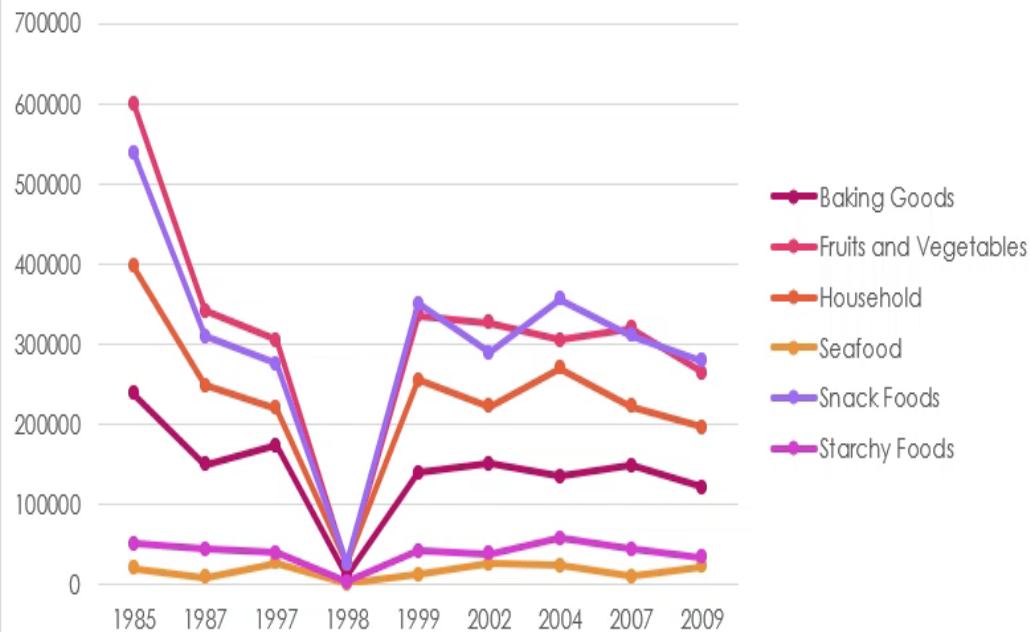
Average items MRP per category in each tier



Number outlets established per year



Total sales for each category per year



Item_Type

Breakfast

Canned

Dairy

Frozen Foods

Fruits and V...

Hard Drinks

Health and ...

Household

Meat

Others

Seafood

Snack Foods

Soft Drinks

Starchy Foods

Location

Tier 1

Tier 2

Tier 3

Insights & Business Impact

- **Expand in Tier 3 Cities:** Tier 3 locations generate the highest overall sales, indicating strong market demand and growth potential. Future outlet expansion should prioritize these regions.
- **Product Focus for New Outlets:** Emphasize vegetables, fruits, snack foods, and household items, as these categories consistently outperform others in sales volume and customer preference.
- **Pricing Strategy:** Baking goods show lower prices compared to other categories – an opportunity to enhance pricing and profit margins while maintaining demand.
- **Low-Performing Categories:** Avoid overstocking or expanding seafood and starchy food in Tier 3 cities due to weak sales performance.
- **Revenue Optimization:** Analysis shows a positive correlation between price and sales – suggesting that strategic price increases could lead to higher overall revenue without significantly affecting demand.